

When to rerank tool retrieval

A component-disjoint empirical study
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Held-out queries	Fixed tool corpus	Disjoint gold-tool groups	Full execution check
1,500 / three sources	37,292 tools	1,174 components	1,500 primary decisions

Abstract

The primary utility router meets the predeclared 0.01 absolute nDCG engineering noninferiority criterion. Its nDCG@10 is 0.5273, versus 0.5241 for always-on reranking (difference +0.0032; query 95% CI [-0.0016, +0.0080]; component 95% CI [-0.0016, +0.0079]). Real conditional execution verified 1,149 CE calls and 22,980 scored pairs across all 1,500 queries, a 23.4% call reduction.

The simpler disagreement gate has a stronger observed quality point estimate: nDCG@10 0.5314 with 1,077 required calls, versus 0.5273 with 1,149 for the primary router. Exploratory learned-minus-disagreement and learned-minus-source-matched-random component intervals both include zero; learned allocation superiority is not established.

On 90 source-balanced confirmation queries with three repetitions per policy, measured warm sequential CPU mean retrieval time is 791.4 ms for the primary router and 920.7 ms for always-on reranking. The observed mean-time reduction is 14.0% (paired query-bootstrap 95% CI [8.2%, 20.0%]). Initialization, network, queueing, and concurrent serving are excluded.

Question and declared contribution

Can cheap first-stage ranking signals predict when tool reranking is worth its compute? This study transfers a known learned-gating design to tool retrieval and tests it using a strict supervised-router holdout, fixed budgets, matched random allocation, full conditional-execution checks, and repeated actual CPU timing.

Relation to existing work

ToolRet supplies heterogeneous tasks/documents [1]; ToolRerank studies adaptive truncation and hierarchy [2]. Learned feature gating, held-out selection and matched controls already appear in Bacellar [3]; Lookahead-R studies budget-aware tool planning [4]. This study contributes transfer, execution evidence and diagnostics, rather than a new gating algorithm.

The complete result is reported under its frozen scope, including negative or inconclusive outcomes. No accepted publication, faculty supervision, or cross-paper state-of-the-art claim is implied.

1. Frozen design and supervised-router holdout

Partition	Queries	Gold components	Permitted use
Observed development	300 (100/source)	252	Sole supervised fit and standardization
Fresh calibration	300 (100/source)	249	Prediction quantiles only; no labels for fit/choice
Untouched confirmation	1,500 (500/source)	1,174	Frozen policies, metrics, runtime and diagnostics

The 3,099 eligible queries form transitive shared-positive-tool groups. Pilot-touching groups anchor 929 queries: 300 train and 629 others are quarantined. Remaining whole groups are assigned by fixed hash/quota, reserving 370 queries.

The three partitions share no query IDs, normalized texts, positive tool IDs, or components. Labels define groups, not performance selection. These are not verified API families; held-out tools remain searchable.

Fixed first-stage features

Seven scalar features	Definition
Top-1 disagreement	Indicator that BM25 and dense top results differ.
Jaccard overlap, top 10 / top 20	Intersection / union of IDs at each cutoff.
Reciprocal-weighted overlap, top 20	Sum of minimum reciprocal-rank weights / sum of maximum weights; absent weight zero.
Rank coherence, top 20	1 - mean absolute rank difference / 20; missing position 21.
Query length	$\log_{1p}(\text{whitespace token count})$.
Normalized RRF margin	Top-1 to top-2 score gap / top-1 score, from both top-100 lists.

Fitting and calibration

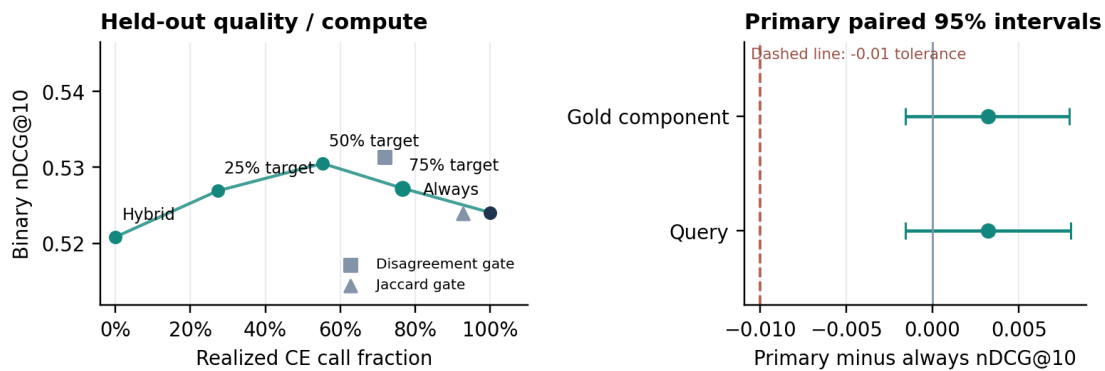
Ridge(alpha 10) predicts development nDCG(always)-nDCG(hybrid). Means/stds use development only; the intercept is unpenalized. Calibration-score quantiles fix 25/50/75% target thresholds. Route iff prediction > threshold; ties skip. Confirmation call fractions can differ.

BM25/base MiniLM top 100 each; equal RRF k=60; CE top 20 only, unchanged tail. Pair length 256; CPU threads 4, interop 1, batch 64. Inference features exclude source/query IDs, lexical tool-ID content, qrels and CE outcomes.

2. Untouched confirmation: quality and compute

Binary relevance at cutoff 10; source means have equal 1/3 weights. Counts are policy-required CE work; primary counts were also verified by real conditional inference.

Policy	nDCG@10	MRR@10	Recall@10	All-label coverage	CE calls	CE pairs
BM25	0.4689	0.4725	0.5840	0.4887	0	0
Base MiniLM	0.4602	0.4583	0.5911	0.5027	0	0
Hybrid RRF	0.5208	0.5196	0.6522	0.5620	0	0
Always rerank	0.5241	0.5301	0.6499	0.5480	1,500	30,000
Disagreement gate	0.5314	0.5388	0.6538	0.5553	1,077	21,540
Jaccard gate	0.5239	0.5276	0.6513	0.5520	1,391	27,820
Utility 25% target	0.5269	0.5272	0.6549	0.5633	409	8,180
Utility 50% target	0.5305	0.5339	0.6569	0.5620	829	16,580
Utility 75% target	0.5273	0.5327	0.6520	0.5527	1,149	22,980



Sole primary hypothesis: utility 75% versus always

Paired bootstrap	Difference /95% interval	Lower bound > -0.01
Source-stratified query	+0.0032 [-0.0016, +0.0080]	True
Source-stratified gold component	+0.0032 [-0.0016, +0.0079]	True

Primary criterion met. Both lower bounds must exceed -0.01 and CE calls must be fewer. The engineering tolerance is not universal equivalence or a power guarantee.

Paired percentile procedures: 10,000 draws, seed 20261003. Component draws preserve query means through per-source ratios. Smaller budgets, controls and sources are exploratory. All-label coverage concerns qrels, not mandatory execution tools.

3. Allocation controls and secondary tradeoffs

The simpler disagreement gate has a stronger observed quality point estimate: nDCG@10 0.5314 with 1,077 required calls, versus 0.5273 with 1,149 for the primary router. Exploratory learned-minus-disagreement and learned-minus-source-matched-random component intervals both include zero; learned allocation superiority is not established.

Random controls distinguish reduced compute from informed allocation. Twenty seeds 0-19 match exact realized call counts. Global controls match total calls; source-matched controls also preserve call counts in each source. The primary policy is not chosen from these comparisons.

Random allocation control	Calls/seed	Mean nDCG	Full seed range
Global / fixed gate	1077	0.5225	[0.5173, 0.5278]
Global / utility 25	409	0.5214	[0.5137, 0.5272]
Global / utility 50	829	0.5216	[0.5165, 0.5276]
Global / utility 75	1149	0.5225	[0.5182, 0.5283]
Source matched / fixed gate	1077	0.5206	[0.5126, 0.5277]
Source matched / utility 75	1149	0.5242	[0.5186, 0.5274]

Exploratory paired differences

Policy minus per-query 20-seed mean	Delta nDCG	Query 95% CI	Component 95% CI
Primary minus global random	+0.0048	[+0.00003, +0.00960]	[+0.00020, +0.00933]
Primary minus source-matched random	+0.0031	[-0.00150, +0.00779]	[-0.00138, +0.00756]
Fixed gate minus global random	+0.0088	[+0.00404, +0.01378]	[+0.00384, +0.01373]
Fixed gate minus source-matched random	+0.0108	[+0.00576, +0.01596]	[+0.00552, +0.01596]

Paired intervals resample queries/components after averaging random-policy outcomes within query. Seed ranges describe allocation randomness, not uncertainty on new queries. These secondary comparisons do not replace the prespecified noninferiority hypothesis or establish multiple-testing-adjusted superiority.

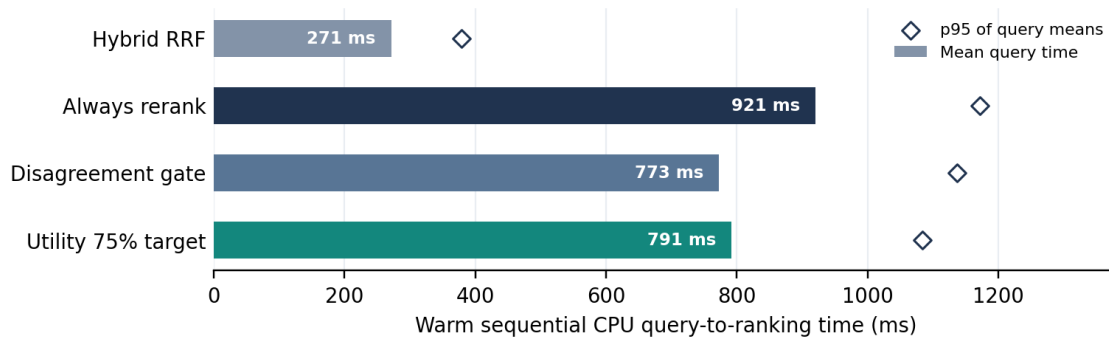
Label-aware budget ceiling: diagnostic only

Realized budget from	Matched calls	Oracle nDCG@10
Disagreement gate	1077	0.5929
Utility 25% target	409	0.5929
Utility 50% target	829	0.5929
Utility 75% target	1149	0.5918

The oracle allocates CE calls to the largest true confirmation-query gains. It uses test labels and reranking outcomes unavailable to the deployed router. It bounds attainable quality for this exact-call allocation problem; it is neither a usable policy nor a competitive result.

4. Actual conditional execution and CPU runtime

On 90 source-balanced confirmation queries with three repetitions per policy, measured warm sequential CPU mean retrieval time is 791.4 ms for the primary router and 920.7 ms for always-on reranking. The observed mean-time reduction is 14.0% (paired query-bootstrap 95% CI [8.2%, 20.0%]). Initialization, network, queueing, and concurrent serving are excluded.



Policy	Mean ms	p50 query mean	p95 query mean	Actual CE calls / requests
Hybrid RRF	271.1	242.1	378.4	0 / 270
Always rerank	920.7	907.0	1172.3	270 / 270
Disagreement gate	772.5	847.7	1137.3	210 / 270
Utility 75% target	791.4	850.9	1083.7	219 / 270

Full-cohort primary verification

Across 1,500 queries, actual execution made 1,149 CE calls and 22,980 scores, skipping 351 calls. Every route and final ranking matched frozen/evaluated expectations. First-stage rankings are cached in this check; end-to-end timing is separate.

Measurement and uncertainty

Hash-selected 30 queries/source, three repeats and four policies give 1,080 requests. Query order is randomized; policy positions are cyclically balanced. Timers include BM25, dense encoding/full search/sort, RRF, router features/prediction, and conditional CE/prefix sorting.

Loaded models/index/embeddings remain in memory. Loading, warmup, validation, network, queues and concurrency are excluded. Repeats are averaged within query before paired source-stratified bootstrap. p50/p95 describe query means; separate event-level percentiles remain in raw evidence.

Runtime: AMD EPYC 9V74 80-Core Processor; Torch threads 4, interop 1; Python 3.12.14. Single-machine warm sequential behavior does not establish production serving speed.

5. Source behavior and failure diagnostics

Source	Groups	Hybrid	Always	Fixed gate	Primary	Primary calls
APIGen	353	0.5851	0.6145	0.6225	0.6169	472
ToolBench	341	0.3588	0.3204	0.3333	0.3301	374
ToolACE	480	0.6186	0.6372	0.6383	0.6348	303

All source-specific results are exploratory. Aggregate performance can conceal source regressions; no source is dropped and no budget is reselected using these outcomes.

Candidate scope and judgment risk

Source	Label recall@20	No gold in top 20	CE wins/losses/same	Truncated gold pairs	Unjudged cross-source top 1
APIGen	0.8010	65	130 / 106 / 264	17 / 481	187
ToolBench	0.5671	132	116 / 184 / 200	70 / 637	330
ToolACE	0.7882	85	118 / 107 / 275	87 / 489	123

The fixed CE prefix cannot recover tools it never scores. Exact pair-tokenizer replay measures query/document clipping, but truncation associations do not identify why a rank changed. An unjudged cross-source tool can still be useful; exact-ID qrels remain unchanged. Name counterparts are a textual warning, not independently established semantic equivalence.

ToolBench always-minus-hybrid nDCG changes by -0.0384; 184 of 500 queries worsen. Of those losses, 152 have no truncated labeled pair. Clipping a positive is therefore not necessary for every loss; this does not identify a cause or rule out clipped distractors. Exact-ID judgment risk and multi-request coverage remain distinct hypotheses.

What the primary router skips

Harmful skips	Beneficial skips	Unchanged skips	nDCG gain forgone / harm avoided
60	87	204	15.740 / 20.542

Harmful means reranking would increase qrel nDCG but is skipped; beneficial means a reranking loss is avoided. Counts use numerical tolerance 1e-12. These are ranking-label effects, not demonstrated execution failures.

Outcome-selected case review

toolbench_query_81 (-0.7359 CE-minus-hybrid nDCG): Transactions and current-month quota are requested. CE promotes an unjudged ToolACE transaction tool; the labeled quota endpoint moves rank 2 to 15. Neither labeled pair is clipped. The primary router skips this rerank. Equivalence and execution are unverified.

toolbench_query_716 (+0.7654 CE-minus-hybrid nDCG): Two labeled Nexweave template tools move from ranks 15 and 13 to 1 and 2. A third positive, the icon-search endpoint, is absent from fused candidates. Ordering improves, while a candidate miss remains outside the reranker's reach.

Cases are deterministically selected by outcome size and query-ID ties. They cannot estimate prevalence or justify a post-confirmation policy change.

6. Evidence chain and reproducibility

Independent audit: **317 checks passed; zero failed or pending.** Data grouping, router fit/thresholds, metric/random/interval results, and runtime evidence are reconstructed independently of the study metric/router/evaluator implementations.

Public prospective protocol commit: 3e813e1af650104f33aaf77febce5a8c1a3f1213. Protocol/data freeze preceded fresh rankings; the model and label-free thresholds were serialized before confirmation quality. Evaluation, full conditional verification and latency use the same router, protocol, corpus, query, and cache fingerprints.

Evidence	SHA-256 prefix (full fingerprint chain in Markdown/sidecar)
protocol	d33ceee4741679c5027b7b55
router	f24613a552d36737f4a17797
evaluation	fda5bd3e67a6990c2e950767
runtime	72b10b04b6527f023637f70c
latency	7ba78367007bc9ab3b5b7e72

Reproduction stages

```
.venv/bin/python scripts/run_confirmation_study.py
--output-dir data/research_reproduction_run1
```

Default: reconstruct pinned source data, check hashes, and freshly evaluate published caches/router. Add --recompute to rerun calibration/fit/rank generation; add --measure-runtime for full conditional CE execution and repeated real timing. Missing embeddings are regenerated.

Verified default replay: numerical results match, and predictions/decisions/per-query/report bytes match exactly. Five live source downloads reconstruct all canonical corpus/cohort fingerprints. Neural-recomputation/runtime flags were not run in this smoke; original-study inference/runtime are separate evidence.

Stage	Entry point under scripts/
Data / pinned rankings	prepare_confirmation_study.py / generate_confirmation_cache.py
Fit / frozen evaluation	fit_utility_router.py / evaluate_confirmation_study.py
Conditional execution / timing	verify_confirmation_runtime.py / benchmark_selective_latency.py
Failure diagnostics	analyze_confirmation_failures.py
Audit / finalization / report	audit_confirmation_study.py / finalize_confirmation_study.py / export_confirmation_report.py

Run from the repository root with PYTHONPATH=src, the recorded research dependencies, CPU Torch wheel, and pinned Hugging Face revisions. The README supplies complete argument examples. Checkpointed embeddings avoid recomputing the corpus; generated evidence refuses unintended overwrite.

Pinned model checkpoints

```
sentence-transformers/all-MiniLM-L6-v2
1110a243fdf4706b3f48f1d95db1a4f5529b4d41
```

```
cross-encoder/ms-marco-MiniLM-L-6-v2
233902d25c440f23af6f7d6e94d2946bac0bee0a
```

Recorded libraries: {'numpy': '2.5.3', 'sentence-transformers': '6.1.0', 'torch': '2.14.1+cpu', 'transformers': '5.18.0'}

7. Interpretation, limitations and references

What this result supports

The primary utility router meets the predeclared 0.01 absolute nDCG engineering noninferiority criterion. Its nDCG@10 is 0.5273, versus 0.5241 for always-on reranking (difference +0.0032; query 95% CI [-0.0016, +0.0080]; component 95% CI [-0.0016, +0.0079]). Real conditional execution verified 1,149 CE calls and 22,980 scored pairs across all 1,500 queries, a 23.4% call reduction.

The study evaluates a fixed router under one retrieval configuration. The primary policy remains fixed even if a secondary policy has a stronger observed point estimate. A failure to establish the tolerance is a completed negative/inconclusive result; it does not authorize tuning on the held-out cohort.

Limits on generalization

Tool-ID components constrain direct supervised label overlap but are not verified API families. All corpus tools remain searchable; this is a router-training holdout, not absent-tool retrieval.

Public pretrained models may have seen related benchmark content. Project-level group separation does not demonstrate pretraining independence.

Whole-group quota assignment and pilot-component quarantine define a particular benchmark cohort, not live-user traffic or a representative API census.

Binary qrels can omit interchangeable tools. All-label coverage, cross-source promotions, and case review do not prove mandatory-tool sets, equivalence, or downstream execution success.

The 0.01 tolerance is a declared engineering tradeoff. Both intervals remain visible. Component resampling covers observed positive-tool sharing, not all semantic dependence; secondary comparisons are exploratory.

Single-machine warm sequential CPU timing excludes loading, networking, queueing and concurrency. No GPU or production service conclusion follows.

Next independently frozen questions

Additional retriever/reranker families, controlled truncation and document-serialization ablations, independently judged tool equivalence, and downstream execution would test how far these observations transfer. None is counted as completed in this report.

References

[1] Shi et al. (2025). Retrieval Models Aren't Tool-Savvy: Benchmarking Tool Retrieval for Large Language Models. Findings of ACL 2025.

<https://aclanthology.org/2025.findings-acl.1258/>

[2] Zheng et al. (2024). ToolRerank: Adaptive and Hierarchy-Aware Reranking for Tool Retrieval. LREC-COLING 2024.

<https://aclanthology.org/2024.lrec-main.1413/>

[3] Bacellar (2026). Per-Query Gating of LLM Rerankers for Multi-Hop Retrieval. arXiv preprint 2609.22880.

<https://arxiv.org/abs/2609.22880>

[4] Wu, Guo and Li (2026). Lookahead-R: Budget-Aware Tool Retrieval via Execution-Centric Planning. ICMR 2026; arXiv 2609.35811.

<https://arxiv.org/abs/2609.35811>

<https://github.com/michaelbawuah/toolret-hybrid-retrieval>